Dog Breed Identification Using Transfer Learning

1. Introduction

• **Project Title:** [Dog Breed Identification Using Transfer Learning]

• **Team Members:**

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Team Size : 3

Team member : Teja Sri Paluri

Team Leader : Pothanapalli Poojitha

Team member : Potnuru Ankitha

2. Project Overview

Artificial Intelligence has become one of the most powerful technologies for solving real-world problems by enabling machines to analyze and understand visual data. Image classification is widely used in fields such as healthcare, security, wildlife monitoring, and animal identification. However, identifying dog breeds manually is a challenging task due to the large number of breeds and similarities in their physical features. This process often requires expert knowledge, and incorrect identification may lead to poor decision-making in areas such as pet care, adoption, and veterinary support.

This project focuses on developing a web-based Dog Breed Identification system using Transfer Learning and deep learning techniques. The system allows users to upload an image of a dog, and the trained model predicts the breed accurately. Transfer learning helps improve model performance by using pre-trained neural networks, which reduces training time and enhances accuracy even with limited data. The model is trained on a Kaggle dog breed dataset and achieves approximately 85% accuracy.

The project also integrates data preprocessing, model optimization, and visualization of prediction results through an easy-to-use interface developed using HTML, CSS, and Flask API. By combining Artificial Intelligence with a user-friendly web application, this solution simplifies the process of dog breed identification. It is useful for pet owners, veterinarians, animal shelters, and researchers. The system demonstrates how modern AI technologies can improve decision-making and enhance user experience in real-world applications.

Purpose:

The purpose of this project is to develop an intelligent and user-friendly system for accurate dog breed identification using Artificial Intelligence and Transfer Learning. The system aims to simplify the process of recognizing dog breeds by allowing users to upload images and receive instant and reliable predictions. This reduces the need for expert knowledge and saves time in identifying breeds, especially for mixed or visually similar dogs.

2. IDEATION PHASE

2.1 Problem Statement

Identifying dog breeds manually is a challenging and time-consuming process due to the large number of breeds and similarities in their physical characteristics. Many pet owners, animal shelters, and veterinary assistants lack the necessary knowledge to accurately recognize dog breeds, especially in the case of mixed breeds. Incorrect identification can lead to improper care, training, and health management.

Existing online tools and applications often provide inaccurate results, require paid subscriptions, or are difficult to use. Additionally, these systems are not accessible to all users and do not provide a simple and reliable solution. Therefore, there is a need for an intelligent and automated system that can accurately identify dog breeds using Artificial Intelligence and deep learning techniques. This system should be easy to use, fast, and accessible through a web-based platform.

2.2 Empathy Map Canvas

Target Users

• Pet owners and dog lovers
• Veterinary professionals and animal health workers
• Animal shelters and adoption centers
• Researchers and students interested in animal recognition
• General users who want quick breed identification

Thinks

• What breed is my dog?
• How can I take better care of my pet?
• Does breed affect behavior and health?
• How accurate are AI-based systems?

Feels

• Curious and excited to know about their dog
• Confused about identifying mixed breeds
• Concerned about pet health and care
• Motivated to use modern technology

Says

• I want a simple and accurate solution
• This application should be easy to use
• AI should help in real-life problems
• I want reliable and fast predictions

Does

• Upload dog images on different apps
• Search online for breed information
• Ask veterinarians or experts
• Compare results from different platforms

Pain Points

• Lack of accurate and reliable tools
• Difficulty identifying mixed breeds
• Time-consuming manual research
• Complex or paid applications
• Low trust in existing solutions

Gains

• Quick and accurate breed identification
• Better understanding of dog behavior and health
• Improved decision-making in pet care
• Easy access to AI-based solutions

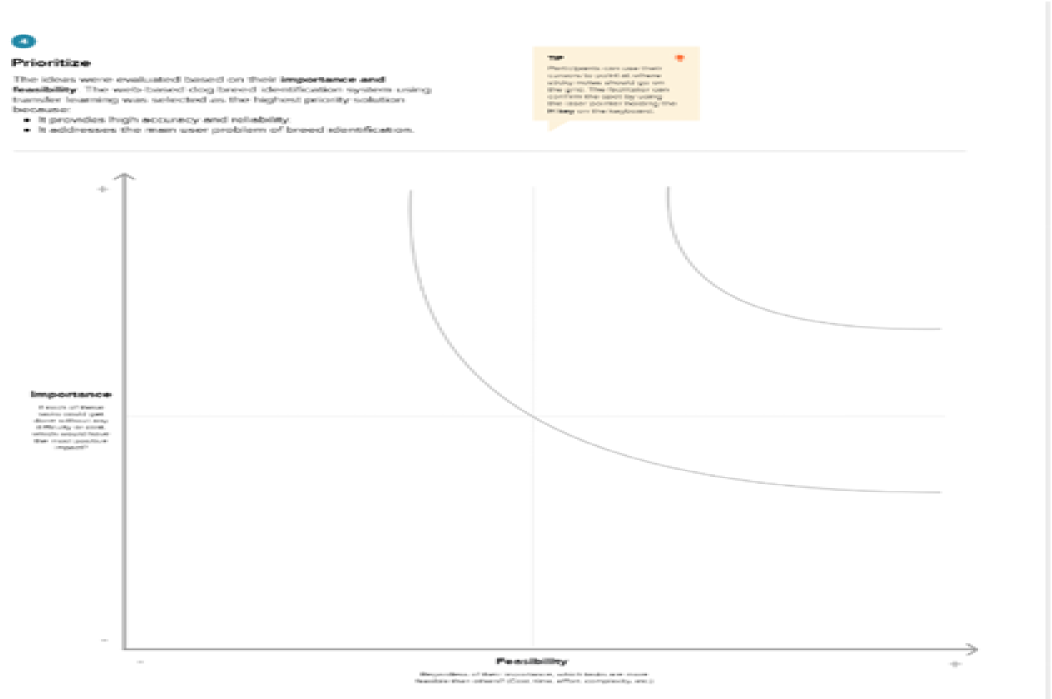
2.3 Brainstorming

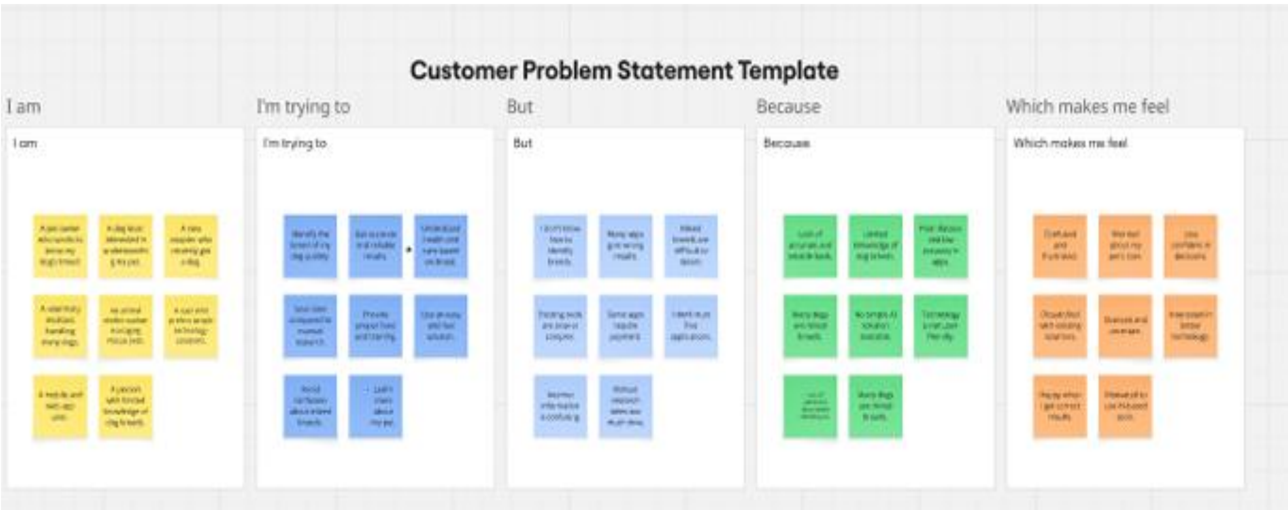
During the brainstorming phase, the team explored multiple innovative ideas to solve the problem of dog breed identification, including:

• Developing an AI-based dog breed prediction system using deep learning
• Creating a web-based platform for easy and quick breed identification
• Using transfer learning to improve model accuracy
• Providing breed details such as health, food, and behavior
• Developing a mobile application for real-time prediction
• Integrating cloud-based deployment for scalability
• Expanding the system to identify other animals in the future

After evaluating feasibility, accuracy, and user requirements, the team selected a web-based dog breed identification system using transfer learning as the most effective and practical solution.

Step-3: Idea Prioritization





3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

1. The user wants to identify the breed of their dog quickly and accurately.
2. The user searches online for dog breed information or mobile applications.
3. The user finds existing tools inaccurate, confusing, or time-consuming.
4. The user accesses the web-based dog breed identification system.
5. The user uploads an image of the dog through the interface.

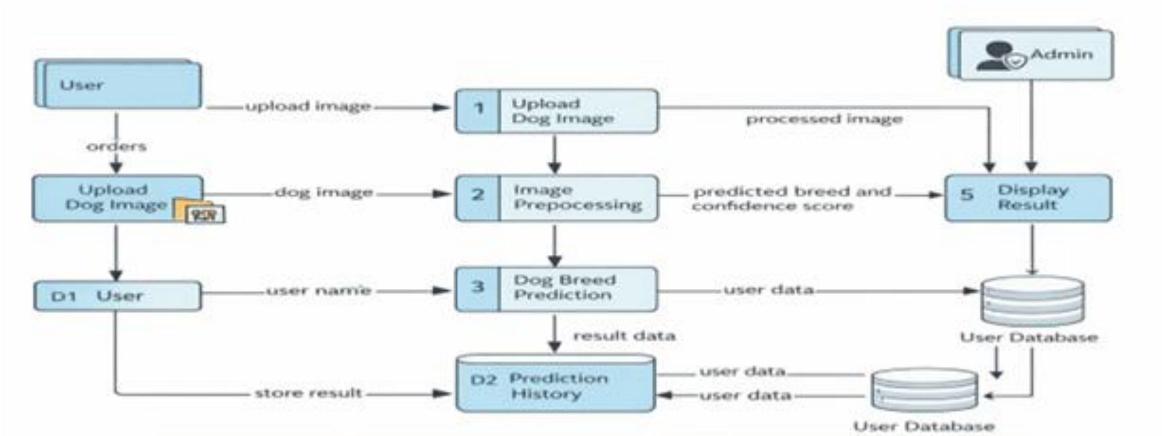
6. The system processes the image and predicts the breed using a trained deep learning model.
7. The user views the prediction result and understands the breed characteristics.
8. The user makes informed decisions related to pet care, adoption, and training.

This journey improves user experience by providing fast, reliable, and accurate breed identification.

3.2 Solution Requirements

Functional Requirements

• Allow users to upload dog images through a web interface
• Preprocess images for model compatibility
• Predict dog breed using transfer learning and deep learning models
• Display predicted breed with confidence score
• Store prediction results and user data
• Provide detailed breed information such as characteristics, behavior, and care
• Support login and user management features
• Allow feedback from users to improve system performance



Dog Breed Identification System Using Transfer Learning – DFD Level 0

Non-Functional Requirements

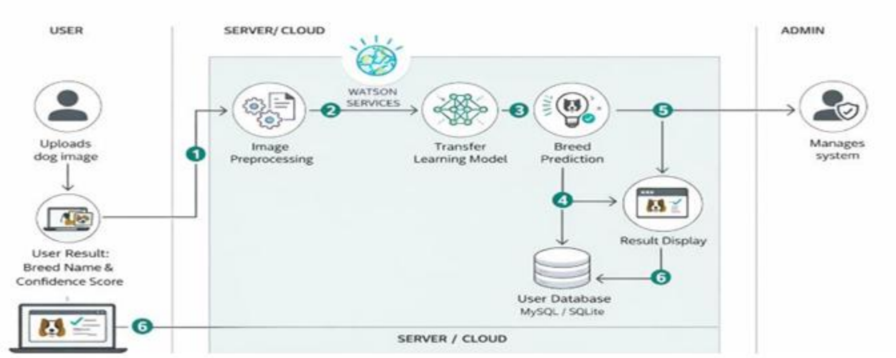
• User-friendly and responsive interface
• Fast prediction and real-time performance
• High accuracy and reliable classification
• Secure data storage and authentication
• Scalable architecture for future cloud deployment
• Compatibility with different devices and browsers

3.3 Data Flow Diagram

The system follows a structured and efficient data flow:

User → Image Upload → Image Preprocessing → Transfer Learning Model → Breed Prediction → Result Display → Database Storage → User

The user uploads an image of a dog, which is preprocessed and analyzed by the trained deep learning model. The system predicts the breed and displays the result. The prediction data is stored in the database for future reference and system improvement.



3.4 Technology Stack

Frontend Development: HTML, CSS, Bootstrap
Backend Development: Python, Flask API
Machine Learning: TensorFlow, Keras, Transfer Learning
Image Processing: OpenCV, NumPy
Database: MySQL / SQLite
Development Tools: VS Code
Dataset: Kaggle Dog Breed Dataset

This technology stack ensures accurate prediction, fast processing, scalability, and an easy-to-use web interface for real-world applications.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How user interacts with application e.g. Web UI, Mobile App, Chatbot etc.	HTML, CSS, JavaScript / Angular Js / React Js etc.
2.	Application Logic-1	Handles user authentication, request processing, and image upload.	Python, Flask
3.	Application Logic-2	Performs image preprocessing such as resizing, normalization, and data transformation.	Python, OpenCV, NumPy
4.	Application Logic-3	Handles classification and prediction of dog breeds.	TensorFlow, Keras, Transfer Learning
5.	Database	Stores user information, login credentials, and prediction history.	MySQL / SQLite
6.	Cloud Database	Used for large-scale deployment and storage in future.	Firebase / AWS RDS
7.	File Storage	Stores uploaded dog images temporarily before prediction.	Local File System
8.	External API-1	Used for authentication and secure login in future.	IBM Weather API, etc.
9.	External API-2	Purpose of External API used in the application	Aadhar API, etc.
10.	Machine Learning Model	Predicts dog breed from uploaded images using deep learning.	Pre-trained CNN (MobileNet / ResNet)

4. PROJECT DESIGN

The project addresses the challenge of manually identifying dog breeds, which is difficult due to the large number of breeds and similarities in their physical characteristics. Many users lack expert knowledge, leading to incorrect identification and improper pet care. The proposed solution transforms complex deep learning models into an easy-to-use web-based system that automates breed recognition. This enables users to quickly and accurately identify dog breeds and make informed decisions related to pet care, training, and adoption.

The system uses Artificial Intelligence and Transfer Learning to analyze dog images and predict the breed with high accuracy. By integrating advanced machine learning techniques with a simple user interface, the solution improves usability, accessibility, and reliability. It also reduces the time and effort required for manual research and ensures consistent results.

4.2 Proposed Solution

The proposed system is a web-based dog breed identification platform that allows users to upload an image of a dog through a simple interface. The system processes the uploaded image and uses a trained deep learning model based on transfer learning to classify the dog breed. The predicted result is displayed along with confidence levels.

The solution is developed using HTML and CSS for the frontend and Python with Flask API for backend communication. The deep learning model is implemented using TensorFlow and Keras. The system is trained using the Kaggle dog breed dataset and achieves approximately 85% accuracy. This makes the solution reliable and suitable for real-world use by pet owners, veterinarians, and animal shelters.

4.3 Solution Architecture

The architecture of the proposed system follows a layered approach to ensure scalability, flexibility, and efficient data flow:

• **User Interface Layer:** A web-based interface developed using HTML and CSS that allows users to upload dog images and view prediction results.

• **Application and Processing Layer:** The Flask API handles user requests, image preprocessing, and communication between the frontend and the deep learning model.

• **Machine Learning Layer:** A transfer learning model based on pre-trained convolutional neural networks processes the image and predicts the dog breed.

• **Data Storage Layer:** The system stores user information, prediction history, and model results in a structured database.

• **Deployment Layer:** The system can be deployed on local servers or cloud platforms for scalability and availability.

This architecture ensures accurate prediction, efficient processing, secure data management, and scalability for future enhancements.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was developed using an agile and structured approach to ensure systematic progress and timely completion. The development process was divided into four major sprints, each focusing on specific tasks and objectives.

Sprint 1: Dataset Collection and Preprocessing
In this phase, the Kaggle dog breed dataset was collected and analyzed. Image preprocessing techniques such as resizing, normalization, and augmentation were applied to improve model performance. This step ensured the quality and consistency of the input data.

Sprint 2: Model Development and Training
In this phase, a transfer learning model was selected and implemented using pre-trained deep learning architectures. The model was trained and validated to achieve optimal accuracy. Hyperparameter tuning and optimization techniques were used to improve the performance of the model.

Sprint 3: System Development and Integration
During this phase, the web-based user interface was developed using HTML and CSS. The backend was implemented using Python and Flask API. The trained model was integrated into the system to enable real-time prediction and communication between the user interface and the deep learning model.

Sprint 4: Testing, Evaluation, and Documentation
In this phase, the system was tested using various dog images to evaluate accuracy and performance. Functional and performance testing were conducted, and documentation was prepared. This ensured the reliability and usability of the system.

This structured agile approach helped in efficient development, continuous improvement, and successful completion of the project within the timeline.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The developed system was tested to ensure accuracy, efficiency, and reliability in real-world scenarios. The following testing activities were performed:

• The system successfully accepts and processes various dog images uploaded by users.
• Image preprocessing ensures compatibility with the deep learning model.
• The transfer learning model provides accurate breed predictions with approximately 85% accuracy.
• The prediction results are displayed clearly and quickly to the user.
• The web interface is responsive and user-friendly across different devices and browsers.
• The system maintains fast response time and smooth performance during multiple requests.
• The model generalizes well to different dog images and real-time use cases.
• The application ensures secure and stable communication between frontend and backend.

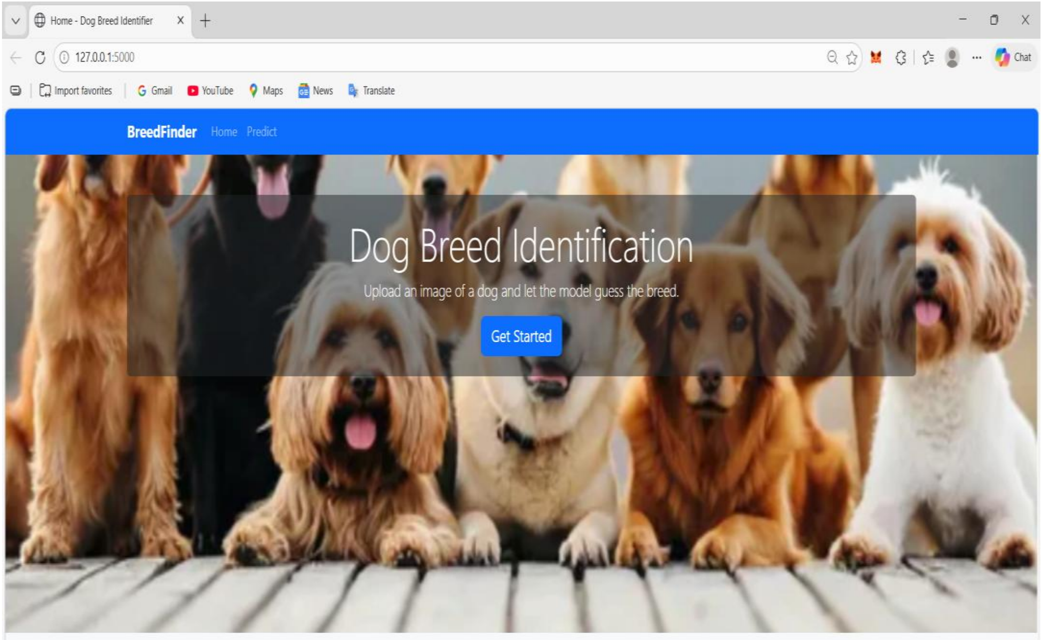
The testing results demonstrate that the system meets functional and performance requirements and is suitable for practical use.

7. RESULTS

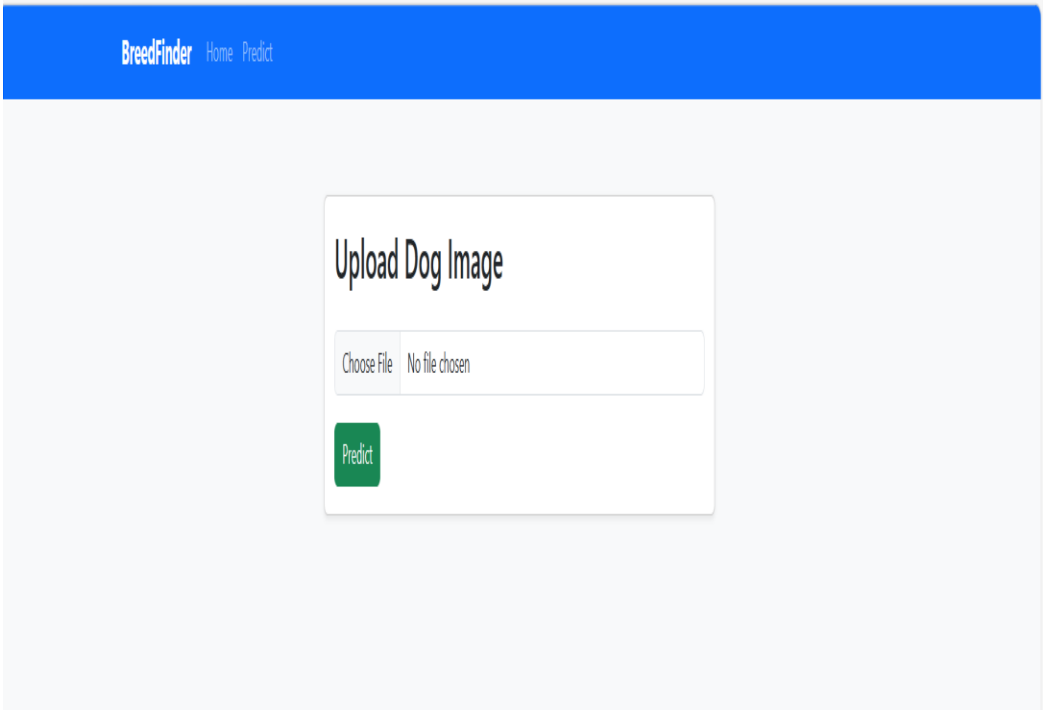
The developed system successfully predicts dog breeds from uploaded images using a transfer learning model. The user interface allows image upload, processing, and real-time prediction. The following outputs were generated:

- The system accepts dog images through a simple and user-friendly web interface.
- Uploaded images are processed and resized before prediction.
- The trained deep learning model predicts the dog breed accurately.
- The prediction result is displayed clearly along with the uploaded image.
- The system achieved approximately **85% accuracy** on the Kaggle dog breed dataset.
- The response time for prediction is fast and suitable for real-time applications.
- The system works for different dog breeds and images with good reliability.

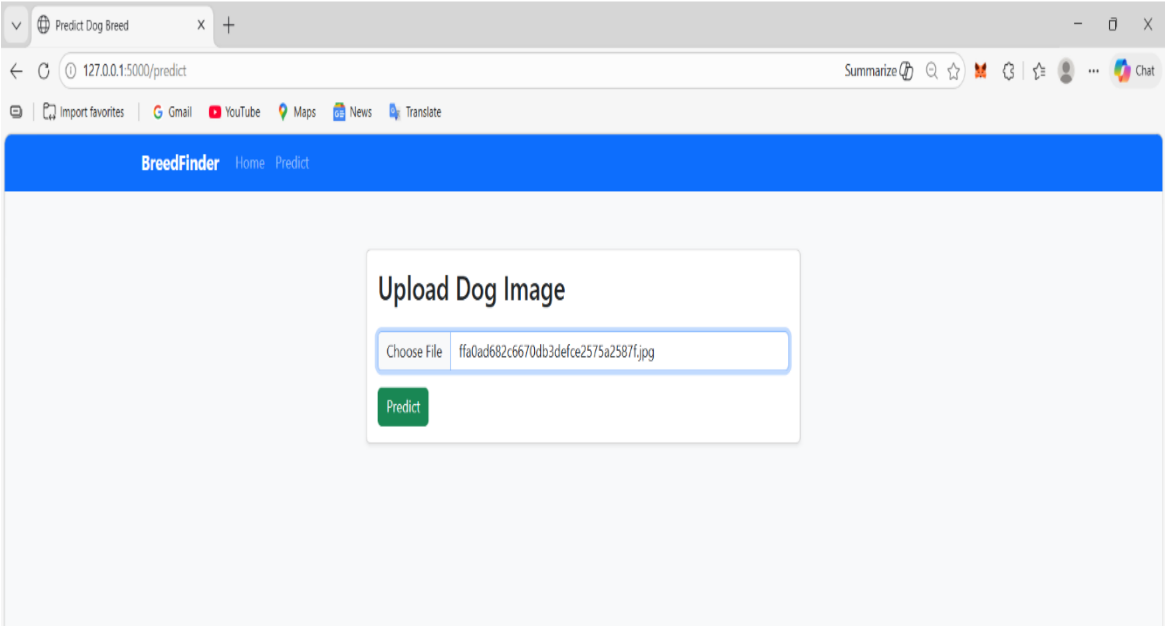
The screenshots below demonstrate the functionality of the system, including image upload, prediction, and result display.



Home - Dog Breed Identifier
127.0.0.1:5000
Import favorites
Gmail
YouTube
Maps
News
Translate
Chat
BreedFinder
Home
Predict
Dog Breed Identification
Upload an image of a dog and let the model guess the breed.
Get Started



BreedFinder
Home
Predict
Upload Dog Image
Choose File
No file chosen
Predict



Predict Dog Breed
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Summarize
Chat
Import favorites
Gmail
YouTube
Maps
News
Translate
BreedFinder
Home
Predict
Upload Dog Image
Choose File
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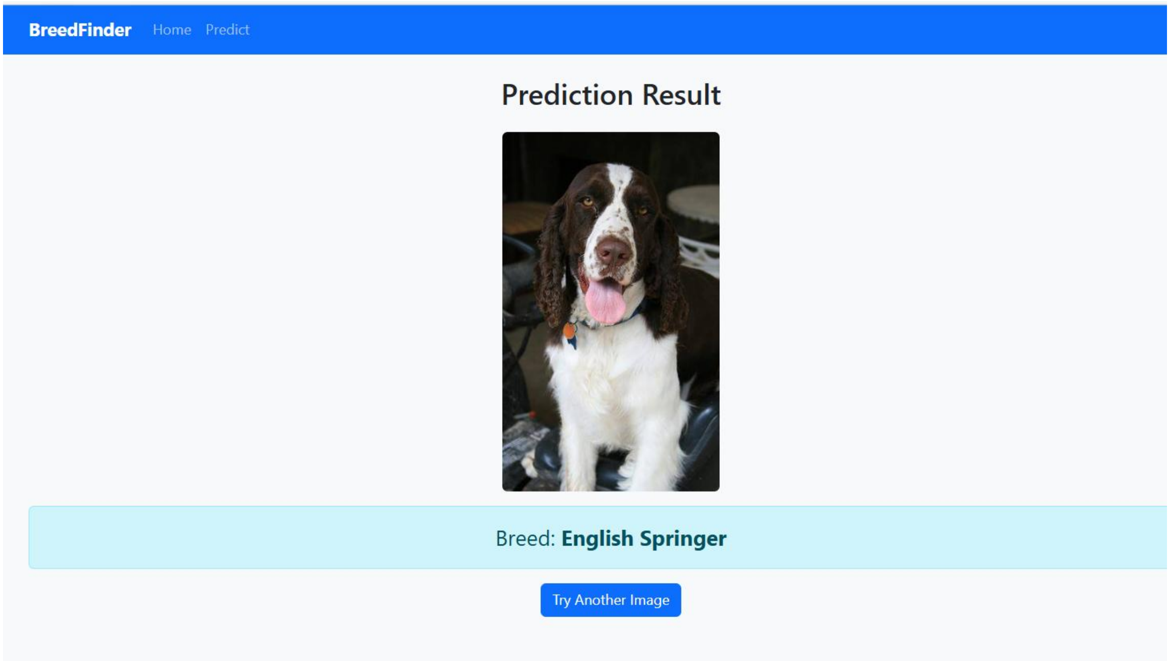
Predict



BreedFinder
Home
Predict
Prediction Result
Breed: English Springer
Try Another Image